CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 2: **Lines and Angles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The minute hand of a clock takes how many minutes to turn through 330°? 1

(a) 60 **(b)** 27 **(c)** 11 **(d)** 55

2. 6 lines are drawn through a point O. The number of straight angles at O (pairs of opposite rays) is: 1

(a) 5 **(b)** 18 **(c)** 7 **(d)** 6

3. Rays OA, OB, OC, OD make equal angles around O (a full turn). The measure of each angle is: 1

(a) 45 **(b)** 55 **(c)** 35 **(d)** 135

4. A wheel has 8 equal spokes, so the angle between two neighbouring spokes is 45°. The angle between spoke 1 and spoke 4 (going one way round) is: 1

(a) 135 **(b)** 90 **(c)** 180 **(d)** 225

5. A clock shows 10 o'clock. After 6 hours, the smaller angle between the clock hands is: 1

(a) 180 **(b)** 120 **(c)** 240 **(d)** 60

6. A wheel turns 9 full turns and then a further quarter turn. The total angle in degrees is: 1

(a) 3240 **(b)** 3330 **(c)** 3420 **(d)** 3510

7. A spinner starts pointing North and turns clockwise 180° then anticlockwise 45°. The final angle clockwise from North is: 1

(a) 135 **(b)** 225 **(c)** 125 **(d)** 145

8. The smaller angle between the hands of a clock at 5 o'clock is 150°. The smaller angle between the hands at 7 o'clock is: 1

(a) 150 **(b)** 210 **(c)** 180 **(d)** 300

9. **Assertion (A):** Four points lie on a line. The number of line segments with end points among these points is 6. 1

Reason (R): Every pair of points gives exactly one line segment.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** At 6 o'clock the angle between the hands of a clock is 180°. 1

Reason (R): At 3 o'clock the angle between the hands of a clock is 90°.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. 10 rays start from O, no two in the same direction, so that every pair makes one angle. Find the number of angles formed. 2

12. Find the smaller angle between the clock hands at 9 o'clock, and the angle the hour hand turns through from 9 o'clock to 11 o'clock. 2

13. A ray rotates 5 full turns and 90° more. Find the total angle turned and the number of right angles in it. 2

14. 12 rays start from O, no two in the same direction, so that every pair makes one angle. Find the number of angles formed. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. At 10 o'clock the hands of a clock make a smaller and a larger angle. Find both and say which one is a reflex angle (if any). 3

16. At 8:00 the minute hand starts turning clockwise and turns through 450°. How many minutes pass, how many right angles does it turn, and what time does the clock then show? 3

17. A pizza is cut by 6 lines through the centre into equal slices (all lines passing through the centre). Find the angle of each slice and the number of slices. 3

OR

At 11 o'clock the hands of a clock make a smaller and a larger angle. Find both and say which one is a reflex angle (if any). 3

18. At 10:00 the minute hand starts turning clockwise and turns through 540°. How many minutes pass, how many right angles does it turn, and what time does the clock then show? 3

OR

A pizza is cut by 9 lines through the centre into equal slices (all lines passing through the centre). Find the angle of each slice and the number of slices. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Rohan has a toy robot that faces north. He makes it turn clockwise through 90° and then through a further 135°. 5

(i) What is the type of the first turn?

(ii) What is the total angle turned?

(iii) What is the type of the total angle?

(iv) How many more degrees must the robot turn to complete one full turn? What fraction of a full turn has it already made?

20. At 11 o'clock, find 5

(a) the angle turned by the hour hand since 12 o'clock,

(b) the angle turned by the minute hand since 12 o'clock,

(c) the smaller angle between the hands,

(d) its type.

OR

A robot starts facing North. It turns clockwise 315°, then 135° anticlockwise, then 90° clockwise. Find its final direction angle clockwise from North and the further clockwise turn needed to face North again. 5

END OF QUESTION PAPER — Total: 40 Marks